

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

LABORATORY OBSERVATION RECORD

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Student Name:	Maddala Surya Prabhath
Roll No:	A24126552094
Branch & Sec:	CSM-B
Date of Execution:	16-08-2026
Course Name:	Full Stack Web Development / Java Programming
Course Code:	23CM4121

EXPERIMENT NO: 02

1. TITLE OF THE EXPERIMENT:

Dynamic Student Profile Generation using JavaScript Classes, DOM Manipulation, and Event Handling

2. MAPPED OUTCOMES:

- Course Outcome (CO): CO1 - Apply JavaScript object-oriented concepts, DOM selection, and event handling techniques to create interactive and dynamic web interfaces.
- Program Outcomes (POs): PO1 (Engineering Knowledge), PO3 (Design/Development of Solutions), PO5 (Modern Tool Usage).

3. AIM:

To design and implement a web application using JavaScript that instantiates a Student class based on user-provided values and dynamically creates, styles, and renders a structured student profile card on the webpage using DOM manipulation upon clicking a button.

4. ALGORITHM / DESIGN:

- Step 1: Create an HTML form interface containing input fields for Name, Roll Number, Department, and CGPA, along with a submit/trigger button.
- Step 2: Define a JavaScript Student class with a constructor accepting parameters: name, rollNo, department, and cgpa.
- Step 3: Attach an event listener (e.g., click) to the submission button.
- Step 4: On button click, retrieve values entered in the input fields using DOM selectors (document.getElementById / querySelector).
- Step 5: Instantiate an object of the Student class passing the retrieved values.
- Step 6: Dynamically construct HTML elements (<div>, <h3>, <p>, etc.) using document.createElement(), assign text content, and apply CSS styles/classes for formatting.
- Step 7: Append the newly created student profile card element to the container DOM node using appendChild() or replaceChildren().

5. TEST CASES & EXECUTION DATA:

TC ID	Input / Request	Expected Result	Status
TC-01	Name: Ravi Kumar Roll No: A23126552001 Department: CSE (AI & ML) CGPA: 8.42	Dynamic DOM card created displaying correct profile details with CSS styling applied.	Passed
TC-02	Name: Maddala Surya Prabhath Roll No: 22A91A05XX Department: CSE CGPA: 9.10	Webpage container updates dynamically and renders the updated student profile card.	Passed
TC-03	Clicking trigger button with missing or empty input fields	Validation logic prompts input or renders default placeholder without throwing unhandled exceptions.	Passed

6. OUTPUT PROOF:

Repository Code & Execution Output Link:

https://github.com/maddalasuryaprabhath/fs_lab/tree/main/Observation/task-3

Rendered Student Profile Output Log:

```
=====
                        STUDENT PROFILE
=====
Name       : Ravi Kumar
Roll No    : A23126552001
Department : CSE (AI & ML)
CGPA       : 8.42
=====
[DOM Tree Output: <div class="profile-card"> appended successfully]
```

7. LEARNING OUTCOME / RESULT:

Successfully implemented and verified the dynamic student profile generator program. Gained practical understanding of JavaScript ES6 classes, object instantiation, DOM element creation (`document.createElement`), modifying node properties (`textContent`, `classList`), handling DOM events (`addEventListener`), and dynamic layout styling using CSS.